

Beyond Fake Effective Sample Size: Normalizing Flow Boltzmann Sampling with Initiative Mode Discovery

Qi Feng¹, Rongjie Lai², Di Qi², and Xuda Ye²

¹Department of Mathematics, Florida State University, Tallahassee, FL 32306

²Department of Mathematics, Purdue University, West Lafayette, IN 47907

May 21, 2026

1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$ known only up to its normalization constant arises across molecular simulation, Bayesian inference, and statistical physics, and is widely tackled with Boltzmann generators built on normalizing flows (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018; Noé et al., 2019; Wu et al., 2020). A flow is an invertible parametric map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ trained so that the pushforward $\nu := G_{\#}^{-1}\mu_0$ of an easily samplable reference μ_0 (typically a Gaussian) approximates μ . The trained flow returns independent samples from ν along with an exact, tractable density, and importance reweighting against μ then converts those into Monte Carlo estimates with controllable finite-sample error.

Two training objectives dominate the literature. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse KL

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (1)$$

which needs only samples from ν and is therefore cheap to estimate. It is, however, mode-seeking: on multimodal μ the flow tends to collapse onto a subset of the modes from which it cannot recover (Noé et al., 2019; Wu et al., 2020). The opposing family minimizes the forward KL (Gabrié et al., 2022; Wirsberger et al., 2020),

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is mode-covering but needs samples from μ . Such samples are produced by annealed importance sampling (AIS) (Neal, 2001), which transports ν toward μ through a ladder of intermediate distributions while accumulating importance weights. The flow-AIS bootstrap (FAB) of Midgley et al. (2022) couples AIS to flow training and replaces the target by the mode-sensitive surrogate μ^2/ν to encourage exploration.

A separate line of work drops the explicit-density requirement and trains a diffusion- or SDE-based sampler directly: the path-integral sampler (Zhang and Chen, 2022), denoising diffusion samplers (Vargas et al., 2023), and the more recent iterated denoising energy matching (Akhound-Sadegh et al., 2024) learn a stochastic process whose terminal distribution matches μ . These samplers are expressive and scale well, but they do not deliver an exact tractable density of their output, so importance reweighting against μ is no longer available. We commit to flow-based methods for that reason: the invertibility of G gives an explicit ν and therefore an exact, finite-sample importance-sampling pipeline, in which an effective sample size (ESS) (Liu, 2001; Doucet et al., 2001) computed from the flow’s own weights is the gold-standard diagnostic of training quality.

The catch is that ESS is gold-standard only when μ and ν are evaluated on the same support. If ν misses modes of μ , the AIS chain initialized at ν rarely escapes the basin ν already covers (Noé et al.,

2019; Midgley et al., 2022), so the AIS samples and the importance weights are computed only against the modes that ν already reaches. The resulting ESS can be close to 1 while a large fraction of the mass of μ is missing entirely. We call this the fake ESS pitfall: ESS alone certifies nothing about mode coverage, and any AIS-reliant scheme inherits the same blind spot.

Our remedy is to augment the forward KL with a family of cross-regularization functionals that penalize variation of the log-ratio $\log(\mu/\nu)$ directly. Let

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

where ω is a non-negative measure on $\mathbb{R}^{2d}$. The motivation is direct: matching ν to μ is the same as flattening $\log(\mu/\nu)$, and the pairwise difference cancels the unknown normalization of μ . As a loss ingredient X_ω has three properties that we will use throughout:

1. **Pointwise control.** The forward KL constrains $\nu \approx \mu$ only in a global Jensen-averaged sense, whereas X_ω penalizes log-ratio variation wherever ω places mass, letting us localize the constraint.
2. **Sample robustness.** X_ω vanishes whenever $\log(\mu/\nu)$ is constant on the support of ω , so its minimizer is insensitive to the precise sampling of ω ; even hand-injected mode locations can be put into ω without corrupting it.
3. **Forward-KL compatibility.** For $\omega = \mu \otimes \mu$ the integrand of X_ω coincides with that of the forward KL up to a sign, so the same autograd graph and the same μ -samples are reused at negligible extra cost; the scales also match and no extra hyperparameter is needed.

Each choice of ω localizes the constraint to a region of $\mathbb{R}^d$; we use three choices with complementary roles. With $\omega = \mu \otimes \mu$,

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (4)$$

which is essentially free by the forward-KL compatibility above. The forward KL controls the μ -mean of $\log(\mu/\nu)$ while X_μ controls its μ -spread, tightening the in-mode density match once ν has reached the modes of μ and reducing the residual bias inherited from imperfect AIS samples.

With $\omega = \hat{\mu} \otimes \hat{\mu}$,

$$X_{\hat{\mu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \hat{\mu}(x)\hat{\mu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (5)$$

where $\hat{\mu}$ is a coarse, deliberately over-spread approximation of μ . By sample robustness the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$, so the sole purpose of $\hat{\mu}$ is to place mass on modes and tails that the current ν has not yet covered; the gradient then pulls the flow toward those regions. We build $\hat{\mu}$ once, before training, with a Quench and Temper (QT) algorithm that diffuses an initial cloud through Brownian noise to bridge basins, runs a deterministic mode-finder to collapse it onto mode centers, and rejuvenates with a short Langevin chain. QT errs on the side of broad coverage and applies uniformly to disconnected modes, long-tailed distributions, and targets supported on low-dimensional manifolds.

With $\omega = \bar{\nu} \otimes \bar{\nu}$,

$$X_{\bar{\nu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \bar{\nu}(x)\bar{\nu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (6)$$

where $\bar{\nu}$ denotes a stop-gradient copy of the current pushforward ν . Sampling from $\bar{\nu}$ is free, since the training loop already pushes a batch from μ_0 through G^{-1} at every step; the detached gradient keeps the loss numerically stable on sharp targets where a true reverse-KL gradient explodes. $X_{\bar{\nu}}$ plays the reverse-KL role without its pathologies: it penalizes mass that ν places outside the support of μ , suppressing the mode-overfitting leakage that X_μ and $X_{\hat{\mu}}$ alone cannot see.

The three weights are complementary: X_μ refines what AIS already reaches, $X_{\hat{\mu}}$ pulls ν toward modes it has missed, and $X_{\bar{\nu}}$ pushes ν away from regions outside the support of μ . Our preferred training loss combines all three with the forward KL,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + X_{(\alpha\hat{\mu}+\beta\bar{\nu}) \otimes (\alpha\hat{\mu}+\beta\bar{\nu})}(\mu\|\nu), \quad (7)$$

where $\lambda, \alpha, \beta \geq 0$ are tunable weights and the third term is X_ω with mixture weight $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$. This mixture form is strictly stronger than the sum $\alpha^2 X_{\hat{\mu}} + \beta^2 X_{\bar{\nu}}$: the pairwise integral also generates cross contributions $|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)|$ with $x \sim \hat{\mu}$ and $y \sim \bar{\nu}$, which anchor the in-support and out-of-support log-ratios to each other. We confirm this empirically: the mixture variant attains both the highest ESS and the cleanest coverage on all of our 2D benchmarks.

We pair ESS with the coverage metric of [Naeem et al. \(2020\)](#) for evaluation, which counts the fraction of QT-discovered $\hat{\mu}$ samples whose k -nearest-neighbor ball is reached by the trained ν . Coverage exposes the modes that a fake ESS hides, so the two metrics together certify both density calibration and mode coverage.

2 Fisher–Rao gradient flow with X functionals

Before turning to the training algorithm, we study the gradient-flow properties of the loss on the space of densities; the convergence rate of this continuous-time idealization is a clean proxy for how trainable the corresponding discrete loss is in practice. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities ([Ambrosio et al., 2008](#); [Villani, 2009](#)) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

In this section we restrict to the loss with only the forward KL and the raw-target regularizer X_μ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (8)$$

where $\lambda > 0$ is a fixed constant. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $\log w(x) = \log \mu(x) - \log \nu(x)$.

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport ([Jordan et al., 1998](#); [Otto, 2001](#)), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery ([Otto and Villani, 2000](#); [Bakry et al., 2014](#)) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao geometry ([Amari, 2016](#); [Bauer et al., 2016](#)), in which the forward KL admits an unconditional exponential decay rate and the regularizer X_μ contributes a qualitatively new, nonlocal birth–death modulation ([Lu et al., 2019](#)) in which the local update at x is driven by how the importance weight at x ranks against the population of weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the Fisher–Rao gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (9)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR flow conserves probability mass automatically.

2.1 Derivation of X_μ -augmented Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν admits the simple form

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (10)$$

so by (9) the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (11)$$

For the X_μ functional, the inner log-ratio difference has first variation

$$\frac{\delta(\log w(x) - \log w(y))}{\delta \nu(x)} = \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of X_μ in x and y ,

$$\begin{aligned}\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \frac{\delta(\log w(x) - \log w(y))}{\delta\nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x)\mu(y) \operatorname{sgn}(w(x) - w(y)) \left(\frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2 \int_{\mathbb{R}^d} w(x) \operatorname{sgn}(w(x) - w(y)) dy,\end{aligned}$$

where the contribution of $\delta(x - y)$ vanishes because $\operatorname{sgn}(w(x) - w(y)) = 0$ at $y = x$. Hence

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu} [\operatorname{sgn}(w(x) - w(y))]. \quad (12)$$

We call S the *rank field* of the importance weight under μ : $S(x) > 0$ where $w(x)$ exceeds the μ -median of w , i.e. where ν undercovers a mode of μ ; $S(x) < 0$ where ν overcovers; and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu} [\operatorname{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\operatorname{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (13)$$

Since the FR gradient is linear in the functional, the FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the direct sum of (11) and (13):

$$\operatorname{grad}^{\text{FR}}[\mathcal{L}](x) = \mu(x) - \nu(x) + 2\lambda\mu(x)S(x).$$

The augmented Fisher–Rao gradient flow $\partial_t \nu_t = -\operatorname{grad}^{\text{FR}}[\mathcal{L}]$ therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\operatorname{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (14)$$

Equation (14) is a *nonlocal* PDE: its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (14) reduces to the pure Fisher–Rao gradient flow of the forward KL.

2.2 Convergence analysis of the X_μ -augmented flow

To analyze the convergence rate of the augmented flow (14), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

Lemma 1 For any probability density ν on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant, i.e. $\nu = \mu$.

Proof. By the definition of S ,

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_\mu [w(x) S(x)] = \mathbb{E}_{x,y \sim \mu} [w(x) \operatorname{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ and averaging the two equal copies gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \operatorname{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation rewrites as

$$\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (15)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \operatorname{KL}(\mu \| \nu_t)$, hence $\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) \leq -\operatorname{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\operatorname{KL}(\mu \| \nu_t) \leq e^{-t} \operatorname{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (16)$$

with no structural assumption on μ — neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts sharply with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (16) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of μ , and producing such samples in high dimensions is itself a hard problem — it is the very reason that AIS is needed, and that the $X_{\hat{\mu}}$ regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (16) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and $\operatorname{KL} + \lambda X_\mu$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_μ is structural rather than rate-based: while the pure-KL FR gradient is local in x , the rank field S_t is global, so the augmented flow (14) couples the update at x to the entire distribution of importance weights under μ . This nonlocal coupling amplifies the birth rate on undercovered regions of μ and suppresses it on overcovered ones, a global comparison that pure forward KL — which knows only whether $\mu(x)/\nu_t(x)$ exceeds or falls below 1 pointwise — cannot produce.

Remark The regularizer X_μ also decreases monotonically along the augmented flow: a direct computation along (14) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality whenever $\nu_t \neq \mu$. Thus X_μ is itself driven to zero in the long-time limit.

2.3 Asymptotic accuracy of biased forward KL training

The convergence analysis of Section 2.2 assumes that the outer expectation in $\text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is taken under the exact target μ . In practice, the algorithm only has access to an approximation $\tilde{\mu}$ of μ supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (17)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target since it is computed from the target potential U and the flow density per sample. Repeating the derivation of Section 2.1 with μ replaced by $\tilde{\mu}$ in the outer integrands gives the Fisher–Rao gradient

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda \tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))].$$

A stationary point ν^* of (17) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) dy \right]. \quad (18)$$

At $\lambda = 0$ the equation collapses to $\nu^* = \tilde{\mu}$: pure forward KL training inherits the bias of the AIS distribution verbatim, and the residual accuracy is exactly $\text{KL}(\mu\|\tilde{\mu})$. For $\lambda > 0$, however, the rank-based correction $1 + 2\lambda \tilde{S}^*$ inflates ν^* above $\tilde{\mu}$ where the importance weight μ/ν^* is high (i.e. where ν^* still undercovers μ) and deflates it where the weight is low, providing a global feedback that pulls ν^* from $\tilde{\mu}$ back toward μ . This is quantified in the following theorem.

Theorem 1 *Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, 1/2]$, so that the bracket in (18) is non-negative and ν^* is a probability density. Any stationary point ν^* of the biased loss (17) satisfies*

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (19)$$

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda \tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda \tilde{S}^*(x) \geq 1 - 2\lambda \geq 0$ since $\lambda \leq 1/2$. The elementary inequality $u \log u \geq u - 1$ for $u \geq 0$ then yields

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma 1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (19). $\square$

The right-hand side of (19) contrasts the two roles of X_μ training. At $\lambda = 0$ the bound is tight at $\nu^* = \tilde{\mu}$ and the residual KL is fixed at $\text{KL}(\mu\|\tilde{\mu})$ — the AIS bias is a hard floor. Switching on the regularizer with $\lambda \leq 1/2$ strictly lowers this floor by an amount controlled by the discrepancy of the importance weights μ/ν^* across the support of $\tilde{\mu}$: the larger this discrepancy, the more the rank-based feedback pulls ν^* toward μ . The X_μ regularizer is therefore not merely a numerical stabilizer for the forward KL flow: it directly improves the asymptotic accuracy of the trained density when the target samples are biased, which is the practically relevant regime.

The threshold $\lambda \leq 1/2$ is a worst-case condition enforcing positivity of ν^* uniformly in x ; in practice the accuracy improvement persists well beyond it. The practical takeaway is that even without the wide-coverage regularizer $X_{\tilde{\mu}}$, adding X_μ alone to the forward KL loss already reduces the bias floor inherited from inaccurate AIS samples.

3 Flow training with X functionals

This section turns the abstract loss of the introduction into a concrete training algorithm for the flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is the easily samplable reference and $\mu \propto \exp(-U)$ is the target. The flow $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is parameterized so that $G_{\#}\mu \approx \mu_0$, and target samples are drawn from $\nu := G_{\#}^{-1}\mu_0$ by importance sampling. The remainder of the section proceeds in four steps: (i) the forward KL together with the AIS pipeline that supplies approximate samples from μ ; (ii) the single-pass computation of X_μ (4) and of the analogously cheap $X_{\bar{\nu}}$ (6); (iii) the Quench and Temper construction of the wide-coverage measure $\hat{\mu}$ used in $X_{\hat{\mu}}$ (5); and (iv) the assembly of all three terms into the total training loss in terms of G . We then close with the ESS and coverage diagnostics used to evaluate the trained flow.

3.1 Forward KL and annealed importance sampling

By the change-of-variables formula, the pullback density of μ_0 under G is

$$\nu(y) = \mu_0(G(y)) |\det J_G(y)|, \quad y \in \mathbb{R}^d, \quad (20)$$

where $J_G(y) \in \mathbb{R}^{d \times d}$ is the Jacobian of G at y . The forward KL loss then reads

$$\begin{aligned} \text{KL}(\mu \parallel \nu) &= - \int_{\mathbb{R}^d} \mu(y) \log \nu(y) dy + \text{const} \\ &= \int_{\mathbb{R}^d} \mu(y) \left(U_0(G(y)) - U(y) - \log |\det J_G(y)| \right) dy + \text{const}. \end{aligned} \quad (21)$$

We deliberately parameterize the loss in terms of G (mapping target to source) rather than G^{-1} , so that no inversion is required during training.

Evaluating (2) requires samples from μ , which are not directly available. The natural surrogate is to draw $y_1, \dots, y_B$ from the pushforward ν and reweight by

$$w(y) = \frac{\mu(y)}{\nu(y)} = \exp \left(U_0(G(y)) - U(y) - \log |\det J_G(y)| \right), \quad y \in \mathbb{R}^d, \quad (22)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ and ν typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of very low quality.

Annealed importance sampling (AIS) bridges this gap by inserting $M + 1$ intermediate distributions

$$\pi_k = \nu^{1 - \frac{k}{M}} (\mu)^{\frac{k}{M}} = \nu \cdot w(y)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M, \quad (23)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each rung of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling:

Algorithm 1: Annealed importance sampling from ν to μ

Input: samples $y_1, \dots, y_B$ from ν , ladder length M

Output: samples $y_1, \dots, y_B$ approximately distributed as μ

```

1 for  $k = 1, \dots, M$  do
2   Compute the per-step weights  $w(y_i)^{\frac{1}{M}}$  on  $y_1, \dots, y_B$ ;
3   Resample  $y'_1, \dots, y'_B$  from the weighted empirical distribution;
   // The resampled set has duplicates; Langevin diffuses them
4   Apply a short Langevin simulation to  $y'_1, \dots, y'_B$ , returning fresh samples  $y_1, \dots, y_B \sim \pi_k$ ;
5 end
```

3.2 Single-pass computation of X_μ and $X_{\bar{\nu}}$

Substituting the explicit log-ratio under the flow map G into the definition (4) of X_μ , the functional becomes

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(y)\mu(y') \left| \left(U_0(G(y)) - U(y) - \log |\det J_G(y)| \right) - \left(U_0(G(y')) - U(y') - \log |\det J_G(y')| \right) \right| dy dy'. \quad (24)$$

The integrand inside the absolute value is exactly the log-ratio $\log(\mu/\nu)$ that appears in the forward KL (2), so a single forward pass through G on a μ -batch produces both losses:

Algorithm 2: Single-pass computation of forward KL and X_μ

Input: batch $\mathbf{y} = (y_1, \dots, y_B) \in \mathbb{R}^{B \times d}$ of samples from μ
Output: forward KL term and X_μ term
 // $\mathbf{z}$ carries the autograd graph through G
 1 Compute $\mathbf{z} = U_0(G(\mathbf{y})) - U(\mathbf{y}) - \log |\det J_G(\mathbf{y})|$;
 // $\mathbf{z}'$ inherits the graph of $\mathbf{z}$ at zero cost
 2 Sample a random permutation σ of $\{1, \dots, B\}$ and set $\mathbf{z}' = \mathbf{z}[\sigma]$;
 3 Return $\text{mean}(\mathbf{z})$ as the forward KL term and $\text{mean}(|\mathbf{z} - \mathbf{z}'|)$ as the X_μ term;

Backpropagation goes through $\mathbf{z}$ exactly once, identical to a vanilla forward-KL step, so X_μ adds essentially no extra training cost.

The same machinery computes $X_{\bar{\nu}}$. Replacing the input batch $\mathbf{y}$ above by a batch $\bar{\mathbf{y}} = (\bar{y}_1, \dots, \bar{y}_B)$ drawn from $\bar{\nu}$ produces

$$X_{\bar{\nu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \bar{\nu}(\bar{y})\bar{\nu}(\bar{y}') \left| \left(U_0(G(\bar{y})) - U(\bar{y}) - \log |\det J_G(\bar{y})| \right) - \left(U_0(G(\bar{y}')) - U(\bar{y}') - \log |\det J_G(\bar{y}')| \right) \right| d\bar{y} d\bar{y}', \quad (25)$$

which is just the second line of the algorithm above with the sample source changed. The batch $\bar{\mathbf{y}}$ is the cheapest available: it is the raw pushforward $\bar{y}_i = G^{-1}(x_i)$ of source samples $x_i \sim \mu_0$ taken under `torch.no_grad`, so $\bar{\nu}$ is treated as a fixed snapshot of the current pushforward for the purposes of gradient computation. The detached sampler keeps the loss numerically stable on sharp targets, where differentiating through the sampling chain—as a true reverse-KL gradient does—would explode.

3.3 Initiative mode discovery in $X_{\hat{\mu}}$ with the QT algorithm

Computing $X_{\hat{\mu}}$ requires samples from the wide-coverage measure $\hat{\mu}$. The samples are prepared once, before training, as a fixed pool of P points $x_1, \dots, x_P$ distributed as $\hat{\mu}$; at each training step a mini-batch of size B is drawn from the pool and refreshed by a short Langevin chain before being fed into the single-pass algorithm above with μ replaced by $\hat{\mu}$. The pool is produced by Quench and Temper (QT), which proceeds in three stages: diffusion (melt), optimization (quench), and rejuvenation (temper).

Algorithm 3: QT: initiative mode discovery

Input: samples $x_1, \dots, x_P$ from μ_0
Output: samples $x_1, \dots, x_P$ forming $\hat{\mu}$
 // Diffusion (melt): scatter the samples to a wider region of $\mathbb{R}^d$
 1 Choose a large $\sigma > 0$ and update $x_i \leftarrow x_i + \sigma \xi_i$ with $\xi_i \sim \mathcal{N}(0, I_d)$ i.i.d.;
 // Optimization (quench): drive each sample toward a mode center of μ
 2 Run LBFGS on U for a fixed number of steps to update $x_1, \dots, x_P$;
 // Rejuvenation (temper): spread the samples around each mode
 3 Run a short Langevin simulation targeting μ on $x_1, \dots, x_P$;

The output samples concentrate near the mode centers of μ , and the initial Brownian diffusion ensures that even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ on which the current pushforward ν places too little mass, driving the flow to discover modes that AIS alone could never reach.

A large diffusion σ biases the QT output toward modes that lie far from the bulk of μ_0 . This is desirable rather than problematic: the modes the current pushforward fails to cover are precisely those far from μ_0 , and the bias is also harmless because sample robustness (Property 2 in the introduction) implies the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$.

3.4 Total training loss

Putting the three terms together, the total loss in terms of the flow map G is

$$\begin{aligned} \mathcal{L}(G) = & \underbrace{\mathbb{E}_{y \sim \mu} [U_0(G(y)) - U(y) - \log |\det J_G(y)|]}_{\text{KL}(\mu \parallel \nu)} \\ & + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu} \left[\left| (U_0(G(y)) - U(y) - \log |\det J_G(y)|) - (U_0(G(y')) - U(y') - \log |\det J_G(y')|) \right| \right]}_{X_{\mu}(\mu \parallel \nu)} \\ & + \underbrace{\mathbb{E}_{y, y' \sim \alpha \hat{\mu} + \beta \bar{\nu}} \left[\left| (U_0(G(y)) - U(y) - \log |\det J_G(y)|) - (U_0(G(y')) - U(y') - \log |\det J_G(y')|) \right| \right]}_{X_{\alpha \hat{\mu} + \beta \bar{\nu}}(\mu \parallel \nu)}, \end{aligned} \quad (26)$$

where the three expectations are estimated on the per-step batches described above: AIS-transported samples for the μ -integrals (forward KL and X_{μ}), and a mixture batch of αB QT samples from $\hat{\mu}$ together with βB stop-gradient pushforward samples from $\bar{\nu}$ for the third term.

Although (λ, α, β) are three free parameters in principle, the X functional is by construction on the same scale as the forward KL, so any choice of order 1 works in practice. Two regimes of the loss (26) are worth highlighting:

1. **Simple regime**, $\lambda = 1$ and $\alpha = \beta = 0$. Only X_{μ} is added on top of the forward KL. This term is free (it reuses the AIS samples and autograd graph of the forward KL) and is enough whenever the source μ_0 already overlaps every mode of μ : the result is a direct accuracy and stability boost over plain forward-KL training, with no extra hyperparameter and no extra sample source.
2. **Mode-discovery regime**, $\lambda = 1$ and $\alpha = \beta = 1/2$. The mixture term is turned on with a balanced half-half draw from $\hat{\mu}$ and $\bar{\nu}$. The $\hat{\mu}$ half enhances mode discovery by pulling ν toward QT-discovered modes that AIS alone cannot reach, while the $\bar{\nu}$ half restrains mode overfitting by penalizing mass that the flow places outside the support of μ . We recommend this as the default whenever μ_0 may not cover all modes of μ , and use it throughout the experiments.

We have not found re-tuning (λ, α, β) away from these defaults to improve results on any of our benchmarks.

3.5 Evaluation metrics: ESS and coverage

The trained flow G induces the distribution ν as an approximation to the target μ . We assess its quality through two complementary metrics: the effective sample size (ESS), which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ has been reached at all.

The first metric is the effective sample size. Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from ν , with importance weights $w(y_i)$ given by (22). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{\left(\sum_{i=1}^N w(y_i) \right)^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (27)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support of ν* , but modes of μ that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS—hence the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of [Naeem et al. \(2020\)](#). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}$ produced by the Quench and Temper algorithm (Algorithm 3). For each $1 \leq i \leq P$, the k -nearest-neighbor radius of x_i is

$$\text{NND}_k(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq k \right\}, \quad (28)$$

i.e., the distance from x_i to its k -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x}) \right\}, \quad (29)$$

i.e., the fraction of points in $\mathbf{x}$ whose k -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}$ has been reached by the flow, while a coverage substantially below 1 exposes precisely the missing modes that the ESS alone cannot see.

4 Construction of adaptive Boltzmann generator

We construct an adaptive Boltzmann generator that determines the AIS temperature ladder on the fly and yields a numerically stable training scheme. As a rule of thumb, for a d -dimensional system an empirical choice of the ladder length M in AIS (Algorithm 3) is $M = d$, which we motivate below with a Gaussian example.

5 Numerical experiments

5.1 Single-step flow in low-dimensions

5.1.1 2D benchmark distributions

We test the training scheme on three 2D targets that each isolate a different failure mode of the forward KL: Himmelblau (four isolated modes), Annulus (a thin one-dimensional ring of minima together with an isolated central minimum), and Chessboard (a Gaussian mixture on 26 of the 32 black squares of an 8×8 grid, with six cells removed to break the chessboard symmetry). The potentials are listed in Table 1.

name	potential $U(x_1, x_2)$	minima
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{1/3}$, $r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$
Chessboard	Gaussian mixture	26 centers

Table 1: Three 2D benchmark targets.

For all three targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$, with $\sigma = 1.0$ on Himmelblau and $\sigma = 0.8$ on Annulus and Chessboard. The flow is a neural spline flow with 6 coupling transforms, 32 bins, and hidden widths (128, 128), defined on the square box $[-L, L]^2$ with $L = 6, 3.6, 5$ for Himmelblau,

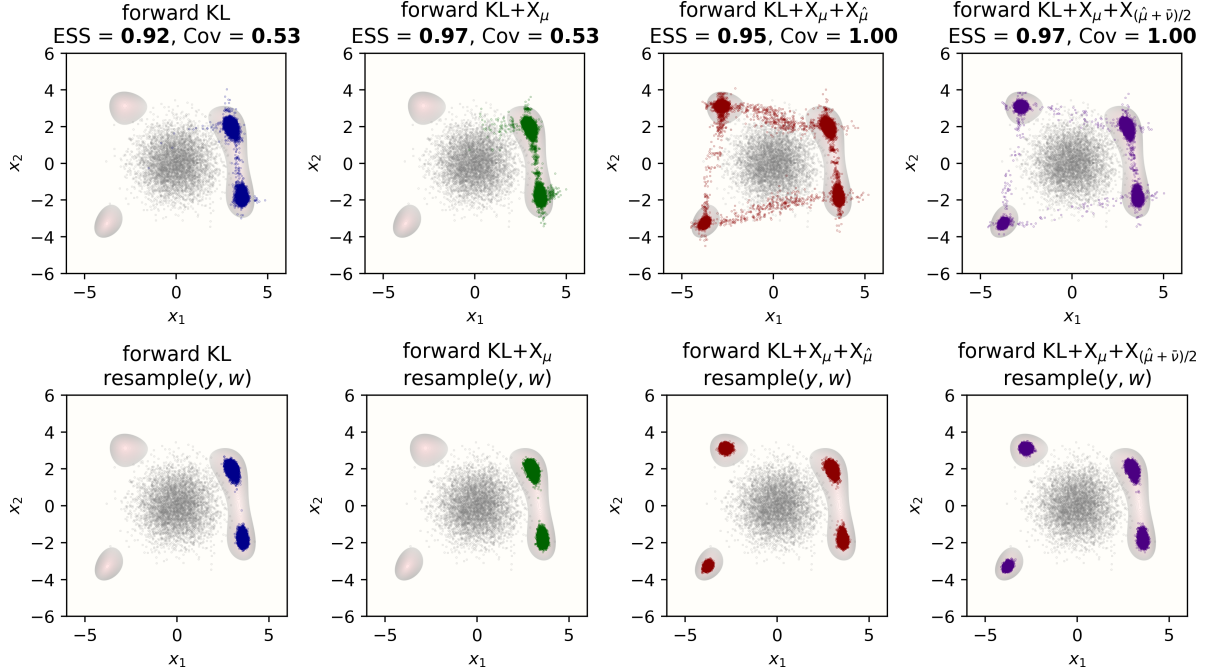


Figure 1: Himmelblau target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns left to right: forward KL, $+X_{\mu}$, $+X_{\mu} + X_{\hat{\mu}}$, $+X_{\mu} + X_{(\hat{\mu} + \bar{v})/2}$. Each pushforward panel title reports the final ESS and the $k = 5$ coverage against $\hat{\mu}$.

Annulus, and Chessboard respectively. Training uses Adam with learning rate 10^{-3} , batch size 500, for 1000 steps; at each step the AIS surrogate is built by one importance-weighted resampling pass against the current pushforward, followed by 50 Langevin steps of size 2×10^{-3} . The loss coefficients are fixed at the preferred default of §3.4, $\lambda = 1$ and $\alpha = \beta = 1/2$. The wide-coverage measure $\hat{\mu}$ is built once before training by applying QT (Algorithm 3) to a fresh batch of 500 source samples, with diffusion standard deviation 2, optimization step 0.5 for 200 LBFGS iterations, and rejuvenation step 2×10^{-3} for 1000 Langevin iterations; the QT pool is then recycled across training steps with one Langevin rejuvenation per step.

For each target we compare four losses obtained by progressively adding one X regularizer at a time:

1. **KL**: the bare forward KL (2);
2. **KL** $+ X_{\mu}$: adding the raw-target regularizer with $\lambda = 1$;
3. **KL** $+ X_{\mu} + X_{\hat{\mu}}$: also adding the wide-coverage regularizer ($\alpha = 1$, $\beta = 0$);
4. **KL** $+ X_{\mu} + X_{(\hat{\mu} + \bar{v})/2}$: the preferred mixture, with $\alpha = \beta = 1/2$.

Himmelblau (mode-coverage regime). The source places nearly all its mass near the central saddle, so the initial pushforward is far from any of the four Himmelblau modes. Forward KL and forward KL $+ X_{\mu}$ both converge to a flow that covers only two of the four modes (coverage ≈ 0.53 in both cases), and the ESS alone gives no warning that the other two modes are missing. Adding $X_{\hat{\mu}}$ steers mass toward the QT-discovered modes and recovers coverage 1.00 (Figure 1). The mixture variant $X_{(\hat{\mu} + \bar{v})/2}$ also reaches coverage 1.00 and attains the highest ESS of the four. This case confirms the role of $X_{\hat{\mu}}$ (or its mixture extension) as the safeguard one needs precisely when μ_0 does not by itself cover all modes of μ — the regime in which the forward KL is silent.

Annulus (shape-correction regime). All four methods reach coverage 1.00, so $X_{\hat{\mu}}$ contributes no mode rescue here and the discriminating quantities are the flow shape and ESS. Forward KL alone settles at ESS ≈ 0.66 , leaving visible axis-aligned filaments across the central barrier. Adding X_{μ} tightens the pushforward to a clean ring-plus-point geometry and lifts the ESS to ≈ 0.96 ; appending $X_{\hat{\mu}}$ on top slightly degrades the

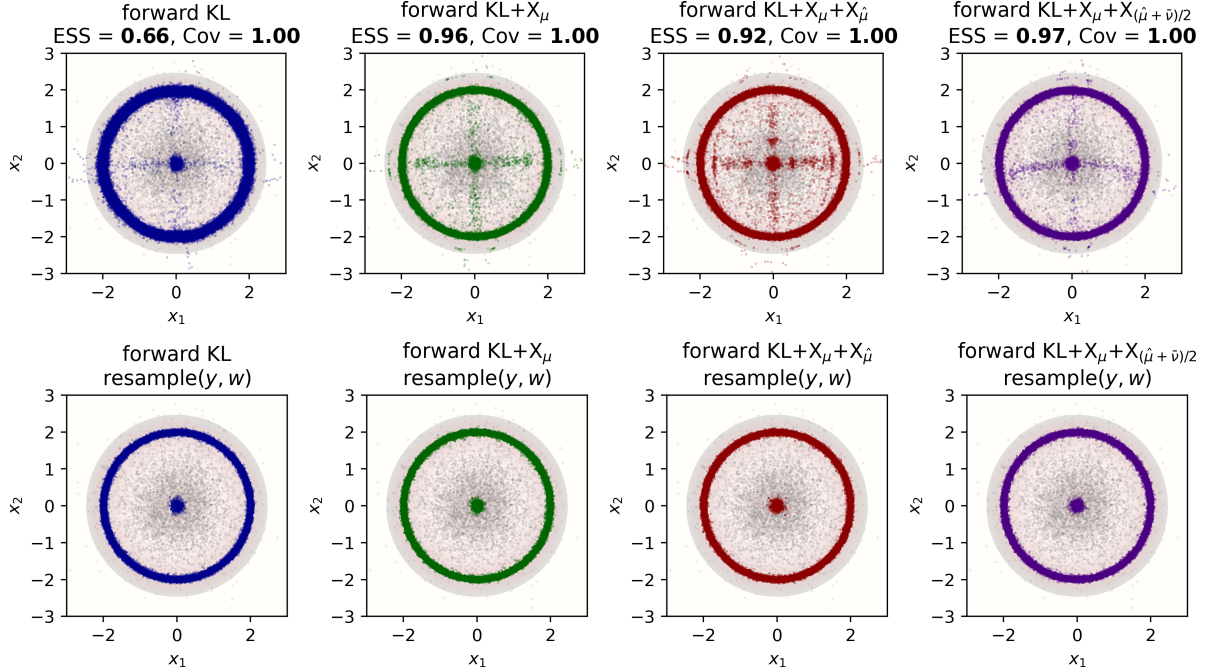


Figure 2: Annulus target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns as in Figure 1.

ESS, while the mixture $X_{(\hat{\mu} + \bar{\nu})/2}$ attains the best ESS of the four (Figure 2). This case isolates the role of X_{μ} as a shape regularizer: it pointwise penalizes log-ratio variations that the global forward KL leaves untouched.

Chessboard (mode-overfitting regime). Coverage is 1.00 for all four methods, but the visible mode structure varies dramatically. Forward KL alone scatters mass loosely across most of the black squares, with some leakage into the six removed cells; adding X_{μ} sharpens each in-cell cluster but does not stop the leakage. Adding $X_{\hat{\mu}}$ on top is actively harmful here: because $\hat{\mu}$ alone is sample-blind to the removed cells, the regularizer pushes mass into every black square indiscriminately and the ESS drops to ≈ 0.64 . The mixture variant $X_{(\hat{\mu} + \bar{\nu})/2}$ resolves this: the $\bar{\nu}$ half of the sampling sees the holes whenever ν leaks mass there, and the resulting cross contributions $|\log(\mu/\nu)(y) - \log(\mu/\nu)(y')|$ with $y \sim \hat{\mu}$ and $y' \sim \bar{\nu}$ anchor in-cell and out-of-cell log-ratios to each other. The trained flow then places mass cleanly on exactly the 26 filled cells and attains the highest ESS of the four (≈ 0.91 , Figure 3).

Across the three targets, the mixture variant $X_{(\hat{\mu} + \bar{\nu})/2}$ is the only one that simultaneously achieves full mode coverage and the highest ESS: it absorbs the mode-discovery role of $X_{\hat{\mu}}$ while also suppressing the mode-overfitting leakage that $X_{\hat{\mu}}$ alone can cause when the target has off-support regions interspersed with its modes.

References

Tara Akhound-Sadegh, Jarrod Rector-Brooks, Avishek Joey Bose, Sarthak Mittal, Pablo Lemos, Cheng-Hao Liu, Marcin Sendera, Siamak Ravanbakhsh, Gauthier Gidel, Yoshua Bengio, Nikolay Malkin, and Alexander Tong. Iterated denoising energy matching for sampling from Boltzmann densities. In *International Conference on Machine Learning*, 2024.

Shun-ichi Amari. *Information Geometry and Its Applications*, volume 194 of *Applied Mathematical Sciences*. Springer, 2016.

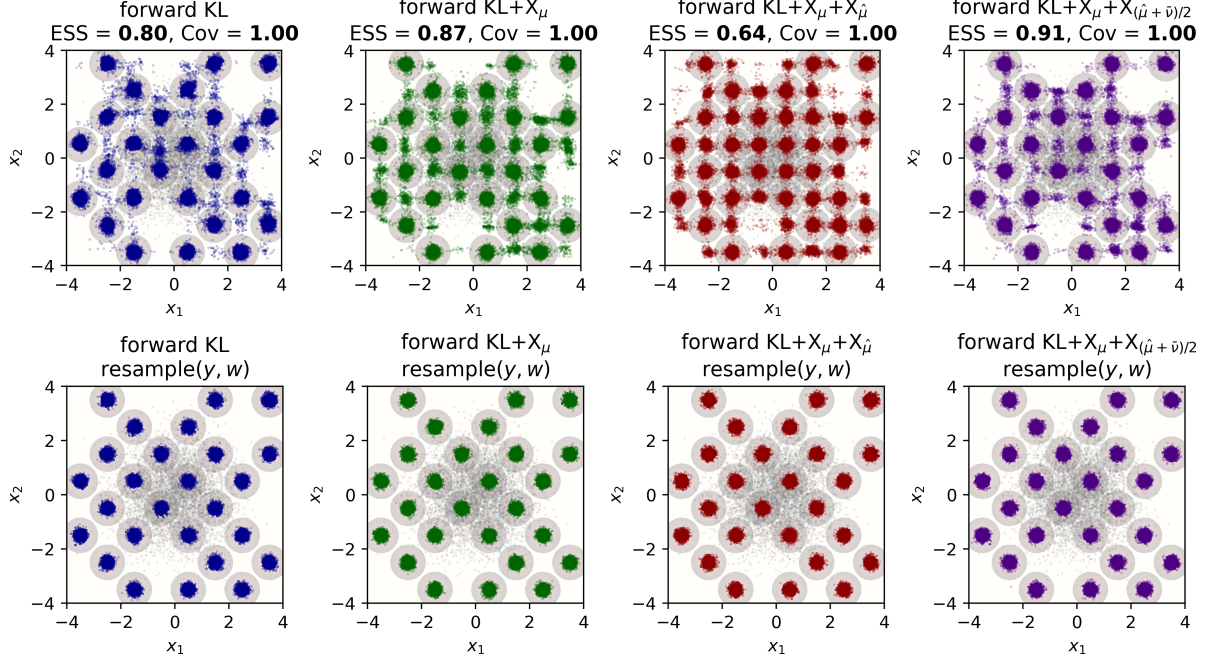


Figure 3: Chessboard target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: resample(y, w) with Langevin rejuvenation. Columns as in Figure 1.

Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, 2 edition, 2008.

Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and Geometry of Markov Diffusion Operators*, volume 348 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2014.

Martin Bauer, Martins Bruveris, and Peter W. Michor. Uniqueness of the Fisher–Rao metric on the space of smooth densities. *Bulletin of the London Mathematical Society*, 48(3):499–506, 2016.

Arnaud Doucet, Nando De Freitas, and Neil Gordon. An introduction to sequential monte carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.

Marylou Gabrié, Grant M Rotskoff, and Eric Vanden-Eijnden. Adaptive monte carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.

Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.

Durk P Kingma and Prafulla Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems*, 31, 2018.

Jun S. Liu. Monte Carlo strategies in scientific computing. 2001.

Yulong Lu, Jianfeng Lu, and James Nolen. Accelerating Langevin sampling with birth–death. *arXiv preprint arXiv:1905.09863*, 2019.

Laurence Illing Midgley, Vincent Stimper, Gregor NC Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. *arXiv preprint arXiv:2208.01893*, 2022.

Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaeyun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.

- Radford M. Neal. Annealed importance sampling. *Statistics and Computing*, 11(2):125–139, 2001.
- Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. *Communications in Partial Differential Equations*, 26(1–2):101–174, 2001.
- Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000.
- George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. *Advances in neural information processing systems*, 30, 2017.
- Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR, 2015.
- Francisco Vargas, Will Grathwohl, and Arnaud Doucet. Denoising diffusion samplers. In *International Conference on Learning Representations*, 2023.
- Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.
- Peter Wirnsberger, Andrew J. Ballard, George Papamakarios, Stuart Abercrombie, Sébastien Racanière, Alexander Pritzel, Danilo Jimenez Rezende, and Charles Blundell. Targeted free energy estimation via learned mappings. *The Journal of Chemical Physics*, 153(14), 2020.
- Hao Wu, Jonas Köhler, and Frank Noé. Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.
- Qinsheng Zhang and Yongxin Chen. Path integral sampler: A stochastic control approach for sampling. In *International Conference on Learning Representations*, 2022.